Project Documentation: AHB-Lite Master System

1. AHB-Lite Standard: What We Implemented vs. What We Ignored

Bus Signal / Feature	Implemented Status	Description / Reason
HTRANS	Partially Implemented	Supported: IDLE (2'b00), NONSEQ (2'b10), and SEQ (2'b11). Ignored: BUSY (2'b01) is not supported.
HBURST	Partially Implemented	Supported: SINGLE (3'b000) and INCR (3'b001). Ignored: Fixed-length bursts (INCR4, INCR8, INCR16) and wrapping bursts (WRAP4, WRAP8, WRAP16).
HSIZE	Partially Implemented	Supported: WORD (32-bit, 3'b010) and HALFWORD (16-bit, 3'b001). Ignored: BYTE (8-bit) and sizes larger than 32-bit.
HPROT	Permanently Disabled	Signal kept in interface for compatibility, but permanently tied to 4'b0000 (No protection control).
HMASTLOCK	Permanently Disabled	Signal kept in interface for compatibility, but permanently tied to 1'b0 (No locked transfers).
HRESP	Single Response Only	Only supports OKAY (1'b0). ERROR responses are not required.
Arbiter / Decoder	Ignored	Point-to-point connection between 1 Master and 1 Slave. No bus matrix or decoder required.

2. Module Architecture & Design Details

2.1. AHB Master (ahb_master.v)

The Master reads transaction requests from the FIFO queue and converts them into standard AHB-Lite protocol signals.

- **Finite State Machine (FSM):** S_IDLE (waiting on FIFO), S_TRANSFER (address phase: NONSEQ for beat 1, SEQ for subsequent beats), and S_LAST_DATA (completing final data phase).
- **Pipelining & Stalling (HREADY):** AHB uses a 2-stage pipeline (Address Phase followed by Data Phase). If hready_i is LOW, the Master freezes all bus signals (HADDR, HTRANS, HWRITE, HWDATA) and pauses FSM state transitions until hready_i becomes HIGH.
- **FIFO Protection:** The current request is NOT popped from the FIFO until the final Data Phase completes successfully on the bus.

2.2. AHB Slave RAM (ahb_slave_ram.v)

The Slave RAM models a 4KB synchronous memory array used to test and verify the Master.

- **Dynamic Wait-State Generation:** Addresses below 0x00000100 complete immediately with 0 wait states (HREADYOUT = 1). Addresses at or above 0x00000100 hold HREADYOUT LOW for 2 clock cycles before completing.
- **Combinational Read Output:** Read data (HRDATA) is driven combinationally during the Data Phase so the Master can capture it on the clock edge ending the transaction.

2.3. Top-Level Interconnect & FIFO (ahb_top.v & fifo.v)

- **fifo.v:** An 8-entry deep FIFO storing the target address, read/write command, write data, transfer size, burst type, and burst length.
- **ahb_top.v:** Connects the FIFO output directly to the Master, and wires the Master bus outputs directly to the Slave RAM inputs.

3. Testbench Verification & Test Cases (tb_ahb_top.v)

The Verilog testbench automatically executes test cases and checks returned read values against an internal scoreboard array.

Test Case 1: Single Word Write & Read (0 Wait States)

- **Goal:** Verify standard 32-bit single word read/write operations without memory stalls.
- **Action:** Write 0xDEADBEEF to 0x00000010, then Read back from 0x00000010.
- **Result:** PASSED (Data returned: 0xDEADBEEF).

Test Case 2: Halfword Write & Read Alignment

- **Goal:** Verify 16-bit halfword accesses (HSIZE = 3'b001).
- **Action:** Write 0x00001234 to 0x00000020, then Read back from 0x00000020.
- **Result:** PASSED (Data returned: 0x00001234 with correct halfword masking).

Test Case 3: Variable Slave Wait States

- **Goal:** Verify that the Master freezes bus signals when the Slave forces wait states using HREADY.
- **Action:** Write 0xCAFEBABE to 0x00000200 (Addr >= 0x100 -> 2 wait states), then Read back.
- **Result:** PASSED (Master held signals stable for 2 additional cycles before capturing 0xCAFEBABE).

Test Case 4: 4-Beat INCR Burst Write & Read (0 Wait States)

- **Goal:** Verify multi-beat incrementing burst transfers (HBURST = INCR, 4 beats).
- **Action:** Write unique values to 0x40, 0x44, 0x48, 0x4C, then issue a 4-beat INCR burst read.
- **Result:** PASSED (Master generated 1 NONSEQ beat followed by 3 SEQ beats with +4 address steps).

Test Case 5: 4-Beat INCR Burst Read with Wait States

- **Goal:** Verify multi-beat burst transfers targeting memory regions that generate dynamic wait states.
- **Action:** Write values to 0x300, 0x304, 0x308, 0x30C, then trigger a 4-beat INCR burst read.
- **Result:** PASSED (Every beat correctly experienced 2 wait states without losing pipeline alignment).

Test Case 6: Back-to-Back Request Queue Pipeline

- **Goal:** Verify continuous execution when requests are queued into the FIFO without gaps.
- **Action:** Push alternating write and read transactions directly into the FIFO back-to-back.
- **Result:** PASSED (Master transitioned smoothly between requests without inserting idle cycles).

4. Verification Results

```
# ================================================================================
#                              VERIFICATION SUMMARY REPORT
# ================================================================================
#    Total Tests Executed  : 6
#    Total Checks Passed   : 13
#    Total Checks Failed   : 0
# --------------------------------------------------------------------------------
#    >>> STATUS: ALL TESTS PASSED SUCCESSFULLY! <<<
# ================================================================================
```